

LOCAL VS COLLEAGUE REPOSITORY COMPARISON

Date: 2026-09-07

Remote repository: panosnaz/infra-automation-platform, branch master, observed HEAD 09b02c8 (feat(aci): Syslog/Fault monitoring policies (ADR-020 Phase G))

Local repository: C:\MCP\INFRA-AUTOMATION-PLATFORM

EXECUTIVE SUMMARY

The two repositories share the same Platform v2 architecture and execution model, but they have diverged in feature focus. The colleague's repository is ahead in platform-wide ACI coverage: RBAC/security domains/local users, COOP/ISIS, and fault/syslog monitoring policies. The local checkout is ahead in the feature work developed during this session: physical/protocol L3Out intent, BGP/OSPF peer and interface modeling, VRF route-leak intent, L4-L7/PBR/service graphs, one-arm ADC graphs, OSPF interface policies, and editable lab templates.

This is a comparison, not a merge recommendation. Remote code was inspected read-only; no fetch, pull, merge, or copy was performed.

SHARED FOUNDATION

Both repositories use the same architectural boundaries:

- Nautobot is the source of truth.
- A generator produces NetAsCode-style YAML.
- Terraform consumes the generated YAML and manages APIC desired state.
- GitLab provides validation, policy, plan, approval, apply, verification, and knowledge capture stages.
- Vault supplies runtime secrets.
- MCP tools write intent to Nautobot rather than directly orchestrating Terraform.
- Unit tests protect transformer/schema/tool behavior.

FEATURE COMPARISON

Area	Local checkout	Colleague remote HEAD
Core Tenant/VRF/BD/EPG	Implemented and tested	Implemented and tested
Contracts/Filters/Subjects	Implemented	Implemented
Logical L3Out	Implemented	Implemented
VLAN pools, Physical Domains, AAEP, IPG	Implemented	Implemented/refined
Leaf profiles/selectors/static paths	Implemented locally in current worktree; plan-validated	
Access-policy work is present in remote history and MCP catalogue		
VMM Domain	Implemented locally; plan-validated	Implemented, with refined domain-binding API
Physical/protocol L3Out	Local implementation includes node/interface profiles, SVI/path attachment, BGP peers, OSPF interfaces, and OSPF policy	Not evident in the inspected remote Terraform/MCP excerpts; verify before porting any related work
VRF route leaking	Local create_vrf_route_leak, generator, aci_vrf_leak_epg_bd_subnet, two-scenario fixture	Not evident in inspected remote excerpts
L4-L7/PBR/Service Graph	Local logical two-arm and one-arm support with PBR bindings	Not evident in inspected remote excerpts
OSPF interface policy	Local tool/resource supports network type, timers, passive flag, auth metadata	Not evident in inspected remote excerpts

RBAC/security domains/local users	Not present in local current implementation	Implemented in remote Phase F
COOP/ISIS/Pod Policy Groups	Not present locally	Implemented in remote Phase E
Fault lifecycle/syslog monitoring	Not present locally	Implemented in remote Phase G
EVPN domain	Implemented locally and live-verified	Implemented and live-verified

DETAILED DIFFERENCE TABLES

PLATFORM AND ARCHITECTURE

Difference	Local checkout	Colleague remote	Assessment
Primary ACI baseline	Platform v2 pipeline plus locally developed ACI extensions	Same Platform v2 baseline, with later ADR-020 phases	Architecture is shared; differences are feature phases, not a competing design
Source of truth	Nautobot objects and JSON Custom Fields	Nautobot objects and JSON Custom Fields	Equivalent principle
Desired-state artifact	Generated NetAsCode-style YAML under platform/netascode/aci/	Same artifact and path	Compatible contract; schema changes must be reconciled
Pipeline ownership	GitLab pipeline owns validation, policy, plan, approval, apply, verification, capture	Same lifecycle	Equivalent
MCP responsibility	Writes intent; does not orchestrate Terraform	Same documented boundary	Equivalent
ACI coverage emphasis	Physical/protocol L3Out, route leaks, L4-L7/PBR, OSPF policy	RBAC, COOP/ISIS, fault/syslog monitoring	Complementary feature priorities
Remote progress marker	Local report date: 2026-09-07	Remote HEAD 09b02c8, Syslog/Fault Phase G, observed 4 days before report date	Remote has newer upstream feature history in platform-wide policy areas

TERRAFORM MODULE DIFFERENCES

Capability	Local Terraform implementation	Remote evidence observed	Difference / action
Tenant/VRF/BD/Subnet	aci_tenant, aci_vrf, aci_bridge_domain, aci_subnet	Same resource family	No material difference identified
EPGs	aci_application_profile, aci_application_epg	Same Phase A family; remote tests also cover EPG domains	Compare relation schema before reconciliation
Contracts/Filters	aci_filter, aci_filter_entry, aci_contract, aci_contract_subject, aci_epg_to_contract	Same Phase A family	No material difference identified
Logical L3Out	aci_l3_outside, external EPG, external subnet	Same logical L3Out baseline	Both require physical/protocol extensions for full routing
VLAN/access objects	VLAN pools, ranges, physical domains, AAEPs, IPGs	Remote has refined Access/Fabric Policy work	Remote may have better append/update semantics; compare exact JSON schemas
IPG policy resources	CDP, LLDP, link-level, spanning-tree, port-security	Remote access-policy work is present	Local added dedicated policy relations and provider enum mappings
Leaf interface path	Leaf profile, access selector, EPG static path	Remote access-policy work exists in history/catalogue	Compare physical path resource semantics before merging
EPG domain binding	Local separate physical/VMM binding paths	Remote consolidated bind_epg_domain API and domain-aware schema	API consolidation opportunity; not a safe mechanical merge
VMM	VMM domain/controller/credential and VLAN-pool relation	Remote VMM with refined	

EPG/domain semantics | Compare credential relation and EPG relation handling |
BGP L3Out	Logical node/profile, fabric node, interface profile, path/SVI attachment, BGP protocol and peer resources	Not evident in inspected remote excerpts	Local feature appears ahead; confirm with full remote file-level diff before merge
OSPF L3Out	OSPF interface profile and typed OSPF policy	Not evident in inspected remote excerpts	Local feature appears ahead
OSPF interface policy	aci_ospf_interface_policy, provider enum translation point-to-point -> p2p, broadcast -> bcast	Not evident in inspected remote excerpts	Local feature appears ahead
VRF route leaking	aci_vrf_leak_epg_bd_subnet with destination VRF, subnet, advertisement flag	Not evident in inspected remote excerpts	Local feature appears ahead; source-VRF provider limitation remains
L4-L7/PBR	Logical device, graph, logical contexts, redirect policies/destinations	Not evident in inspected remote excerpts	Local feature appears ahead in inspected scope
One-arm ADC graph	Dedicated one-arm tool and one-context Terraform path	Not evident in inspected remote excerpts	Local feature appears ahead
COOP/ISIS	Not present locally	Remote typed aci_coop_policy, aci_isis_domain_policy and Pod Policy Groups	Remote ahead; candidate feature to port conceptually
RBAC	Not present locally	Remote AAA/security-domain/local-user Terraform resources	Remote ahead; secrets need careful local adaptation
Fault/syslog monitoring	Not present locally	Remote Phase G singleton policies with destroy-safe handling	Remote ahead; preserve content_on_destroy safety pattern

MCP TOOL AND SCHEMA DIFFERENCES

MCP area	Local checkout	Colleague remote	Difference / action
Core tools	Tenant, VRF, BD, EPG, Contract, L3Out, route leak	Same core plus refined access APIs	Shared baseline
VLAN pool	create_vlan_pool exists in local current worktree	Remote has create_vlan_pool with append-range semantics	Remote API is more mature; compare range merge/idempotency
Physical domain	create_physical_domain	Remote create_physical_domain	Compare defaults and prerequisite validation
AAEP	create_aaep	Remote create_aep naming	Naming/API compatibility issue; choose one canonical public name or alias
IPG	create_leaf_interface_policy_group	Remote same concept	Compare policy relation fields
Leaf profiles/selectors	Local tools exist	Remote access-policy catalogue exists	Compare exact request schema and live inventory validation
Static path	Local create_static_path_binding	Remote access-policy work present	Compare path DN, encapsulation, and update semantics
EPG domain bindings	Separate bind_epg_to_physical_domain and bind_epg_to_vmm_domain	Consolidated bind_epg_domain with domain_type	Remote API is more general; local split API is more explicit
VMM	create_vmm_domain	create_vmm_domain	Remote schema/docs are more mature
L3Out protocols	Local BGP/OSPF L3Out, node/interface, peer, OSPF policy tools	Not evident in inspected remote tools	Local appears ahead
VRF leak	Local create_vrf_route_leak	Not evident in inspected remote tools	Local appears ahead
L4-L7/PBR	Local two-arm and one-arm tools	Not evident in inspected remote tools	Local appears ahead
RBAC/security	Not present locally	create_security_domain, create_local_user	Remote ahead
Status	show_status	show_status and same pipeline-status concept	Shared
Secret boundary	Local avoids raw VMM/OSPF secrets in MCP; uses references/runtime variables		

Remote local-user password also uses Terraform-sensitive runtime input | Same security direction; reconcile secret variable names carefully |

GENERATOR AND INTENT SCHEMA DIFFERENCES

Intent surface	Local checkout	Colleague remote	Difference / action
Location fabric policies	VLAN pools, domains, AAEPs, IPGs, leaf profiles/selectors, VMM, policy settings	Also expanded with COOP/ISIS, Pod Policy Groups, monitoring policies	Remote has broader fabric policy schema
Tenant policy	VRFs, BDs, EPGs, contracts, L3Outs, route leaks, L4-L7 services, OSPF policies	Same baseline plus remote Phase F/G surfaces	Schemas need additive merge, not replacement
EPG custom fields	Physical/VMM bindings and static paths	Remote uses domain-aware `aci_epg_domains` style in inspected tests	This is a likely reconciliation hotspot
L3Out schema	Node profiles, interface profiles, interfaces, BGP peers, OSPF entries	Remote baseline excerpts did not show equivalent full protocol path	Validate before combining
Generator tests	25 transformer tests in latest local focused run	Remote transformer tests cover additional COOP/ISIS, AAA, Pod Policy, syslog/fault keys	Remote has broader generator coverage

TEST AND VALIDATION DIFFERENCES

Test/validation layer	Local checkout	Colleague remote	Difference / action
Transformer unit tests	`tests/unit/test_transformer.py`; 25 passing in latest focused run	Same file covers more remote phases, including AAA and monitoring	Remote has broader current test surface
MCP schema tests	`mcp-server/tests/unit/test_schemas_aci.py`	Same file with more Phase F/access schemas	Remote ahead in catalogue coverage
MCP tool tests	`mcp-server/tests/unit/test_tools_aci.py`	Same file covers VLAN pools, AAEP/IPG, VMM/domain binding, RBAC	Remote ahead in those areas
Registry tests	`mcp-server/tests/unit/test_registry.py`	Same registry/signature checks	Shared quality gate
EVPN tests	Local EVPN transformer/MCP/pyATS-equivalent tests	Remote EVPN tests and live validation	Both have second-domain coverage
Terraform validation	Local `fmt`, `validate`, APIC-backed temporary-state plans for new local fixtures	Remote repository documents broader Phase E/F/G implementation; full local execution not performed from this comparison	Run both suites after reconciliation
Physical L3Out validation	Local plan validated 22 to add, 0 to change, 0 to destroy for combined access/L3Out fixture	Not established from inspected remote excerpts	Local plan evidence exists; apply still requires approval
OSPF policy validation	Local APIC-backed plan validated 23 to add, 0 to change, 0 to destroy	Not established from inspected remote excerpts	Local plan evidence exists
IPG policy validation	Local APIC-backed plan validated 27 to add, 0 to change, 0 to destroy	Not established from inspected remote excerpts	Local plan evidence exists
Live apply status	No apply performed for the newly developed local access/protocol/L4-L7 scope	Remote README/history documents live verification for several earlier domains and newer platform phases	Do not infer apply readiness from plan alone

OPERATIONAL AND DOCUMENTATION DIFFERENCES

| Area | Local checkout | Colleague remote | Difference / action |

|---|---|---|

Local Nautobot endpoint	Isolated instance on port 8081 in current workspace	Remote README contains older port-8080/default credential references in its active-lab section	Use local administration guide, not remote README values
Provider version	Local ACI provider pinned to 2.20.0 and schema-checked locally	Remote uses the same project lineage but version must be checked before merge	Re-run provider schema comparison
Current ADR state	Local ADR-020 includes locally developed Phase D-G style additions from this session	Remote ADR-020/commits contain its own Phase E/F/G progression	Reconcile ADR history manually; do not overwrite decisions
Templates	Local four-file ACI lab templates under knowledge/runbooks/templates/aci/	Not established in inspected remote tree	Keep local templates as scenario-specific input aids
Report confidence	Local counts and plan results verified in this worktree	Remote comparison based on public page/raw files and commit/tree evidence	Full commit-level diff still requires a separately approved read-only clone/fetch process

DIFFERENCE SUMMARY

| Category | Local advantage | Remote advantage |

|---|---|---|

New network-service features	Full physical/protocol L3Out path, route leaks, L4-L7/PBR, one-arm ADC, OSPF policy	None identified in inspected scope
Platform-wide ACI policy	None identified for COOP/ISIS, RBAC, monitoring	COOP/ISIS, Pod Policy Groups, RBAC/security domains/local users, fault/syslog monitoring
MCP API maturity	More detailed L3Out/L4-L7 operations	More consolidated/refined access and RBAC APIs
Generator coverage	Local new service-chain and route-leak intent	Broader platform-policy intent and tests
Validation evidence	Local APIC-backed plans for newly added scopes	Remote documented/live evidence for several established phases

RECOMMENDED RECONCILIATION ORDER

1. Compare remote Phase E/F/G Terraform resources and transformer sections against local main.tf/transformer.py manually.
2. Port concepts, not files: COOP/ISIS, RBAC, and monitoring resources should be adapted to the local isolated-stack paths and secret conventions.
3. Preserve local L3Out protocol, VRF-leak, L4-L7/PBR, OSPF policy, and template work unless remote has a verified newer equivalent.
4. Reconcile MCP naming, especially remote bind_epg_domain versus local split physical/VMM binding tools.
5. Add/merge unit tests before changing generated YAML schema.
6. Run both local test suites and Terraform validation.
7. Run APIC-backed plans with temporary state.
8. Apply only after explicit review and approval.

LOCAL TERRAFORM REFERENCES

Primary module:

- [platform/terraform/aci/main.tf](../../platform/terraform/aci/main.tf)
- [platform/terraform/aci/providers.tf](../../platform/terraform/aci/providers.tf)

- [platform/terraform/aci/variables.tf](../../platform/terraform/aci/variables.tf)

The local ACI module currently contains 52 Terraform resource blocks. Relevant resource families include:

- Core: aci_tenant, aci_vrf, aci_bridge_domain, aci_subnet, aci_application_profile, aci_application_epg
- Policy: aci_filter, aci_filter_entry, aci_contract, aci_contract_subject, aci_epg_to_contract
- Logical L3Out: aci_l3_outside, aci_external_network_instance_profile, aci_l3_ext_subnet
- Access: aci_vlan_pool, aci_ranges, aci_physical_domain, aci_attachable_access_entity_profile, aci_leaf_access_port_policy_group
- IPG policies: aci_cdp_interface_policy, aci_ldap_interface_policy, aci_link_level_interface_policy, aci_spanning_tree_interface_policy, aci_port_security_interface_policy
- Physical access: aci_leaf_interface_profile, aci_access_port_selector, aci_epg_to_domain, aci_epg_to_static_path
- L3Out protocol path: aci_logical_node_profile, aci_logical_node_to_fabric_node, aci_logical_interface_profile, aci_l3out_path_attachment, aci_l3out_floating_svi, aci_l3out_bgp_protocol_profile, aci_bgp_peer_connectivity_profile, aci_l3out_ospf_interface_profile, aci_ospf_interface_policy
- VMM/L4-L7: VMM domain/controller/credential resources, L4-L7 device/interface, service graph, logical contexts, redirect policies/destinations
- VRF leaks: aci_vrf_leak_epg_bd_subnet

The local provider is pinned to CiscoDevNet/aci 2.20.0 in providers.tf and all new resource fields were checked against its local schema.

LOCAL MCP REFERENCES

Primary files:

- [mcp-server/src/mcp_server/tools/aci.py](../../mcp-server/src/mcp_server/tools/aci.py)
- [mcp-server/src/mcp_server/schemas/aci.py](../../mcp-server/src/mcp_server/schemas/aci.py)
- [mcp-server/src/mcp_server/clients/nautobot.py](../../mcp-server/src/mcp_server/clients/nautobot.py)

Local feature families include:

- Core: create_tenant, create_vrf, create_bridge_domain, create_epg
- Policy/L3Out: create_contract, create_l3out, create_vrf_route_leak
- Access: create_vlan_pool, create_physical_domain, create_aaep, create_leaf_interface_policy_group, create_leaf_interface_profile, create_interface_selector, create_static_path_binding, bind_epg_to_physical_domain, bind_epg_to_vmm_domain
- L3Out protocols: create_bgp_l3out, create_ospf_l3out, create_l3out_node_profile, create_l3out_interface_profile, create_l3out_interface, create_bgp_peer, create_ospf_interface, create_ospf_interface_policy
- VMM: create_vmm_domain
- L4-L7: create_l4l7_device, create_service_graph, create_one_arm_service_graph, create_pbr_policy, create_pbr_contract
- Status: show_status

Remote MCP comparison: the inspected remote branch explicitly includes refined create_vlan_pool,

create_physical_domain, create_aep, create_leaf_interface_policy_group, bind_epg_domain, create_security_domain, and create_local_user APIs. Its access schemas use append/update semantics and domain-type-aware EPG bindings. The local names are compatible in intent but should not be blindly copied because the local worktree has additional L3Out/L4-L7 behavior and a different schema evolution.

UNIT-TEST REFERENCES

Local test files:

- [tests/unit/test_transformer.py](../tests/unit/test_transformer.py): generator regression and emitted YAML coverage, including VRF/BD depth, EPGs, contracts, L3Outs, access policies, VMM, L4-L7, and route leaks.
- [mcp-server/tests/unit/test_schemas_aci.py](../mcp-server/tests/unit/test_schemas_aci.py): Pydantic validation for ACI MCP requests.
- [mcp-server/tests/unit/test_tools_aci.py](../mcp-server/tests/unit/test_tools_aci.py): tool-to-client argument/response behavior.
- [mcp-server/tests/unit/test_registry.py](../mcp-server/tests/unit/test_registry.py): catalogue and schema-signature guarantees.
- [tests/unit/test_evpn_transformer.py](../tests/unit/test_evpn_transformer.py) and EVPN MCP tests: second-domain regression coverage.

The last local focused validation recorded during this work was 25 transformer tests and 39 MCP schema/tool/registry tests passing. The local combined access/L3Out plan validated 22 to add, 0 to change, 0 to destroy; the OSPF policy plan validated 23 to add, 0 to change, 0 to destroy; the IPG policy plan validated 27 to add, 0 to change, 0 to destroy.

Remote test references observed in the repository tree and raw files:

- tests/unit/test_transformer.py: broad transformer coverage, including access policies, VMM, COOP/ISIS, Pod Policy Groups, fault/syslog policies, and AAA policy emission.
- mcp-server/tests/unit/test_tools_aci.py: tool-level tests for VLAN pools, Physical Domains, AAEPs, IPGs, VMM, domain binding, Security Domains, and Local Users.
- mcp-server/tests/unit/test_schemas_aci.py: request validation for the expanded remote tool catalogue.
- mcp-server/tests/unit/test_registry.py: MCP registry/signature checks.
- EVPN transformer, MCP, integration, and pyATS-equivalent tests remain present in both project lines.

RECOMMENDED RECONCILIATION

1. Treat the remote branch as the reference for Phase E/F/G ACI coverage: COOP/ISIS, RBAC/security domains/local users, and fault/syslog monitoring.
2. Preserve the local L3Out protocol, VRF-leak, L4-L7/PBR, one-arm service graph, OSPF-policy, and lab-template work unless the remote branch has a newer equivalent after direct file comparison.
3. Compare schema/API changes before any merge. In particular, remote bind_epg_domain is a consolidated physical/VMM operation, while the local checkout currently has separate bind_epg_to_physical_domain and bind_epg_to_vmm_domain tools.
4. Reconcile generated YAML schema changes before combining Terraform modules; do not merge main.tf mechanically.
5. Run both repositories' unit suites and Terraform validation after reconciliation.
6. Perform APIC-backed plans with temporary state before any apply.

LIMITATIONS

This report is based on read-only inspection of the public remote repository page/raw files and the current local worktree. It does not claim a commit-level semantic diff of every file. No remote code was fetched, merged, or copied. The remote README contains older operational values in places; use the local repository's current administration guide and isolated-stack configuration for local credentials and endpoints.